

OPEN ENDED EXPERIMENT – 1

AUTOMATIC HEADLIGHT AND INDICATOR SYSTEM

Aim of the Experiment

To interface a potentiometer as an analog input with the STM32 microcontroller and control three LEDs based on different ADC voltage ranges, simulating an automatic headlight and indicator system.

List of Equipment / Components Required

Sl. No.	Component / Hardware	Specification	Quantity
1	STM32 Microcontroller Board	STM32F103C8T6 (Blue Pill)	1
2	Potentiometer	10 k Ω	1
3	LED	Red / Green / Blue	3
4	Resistor	330 Ω	3
5	Connecting Wires	Jumper Wires	As Required

Theoretical Background

The STM32 microcontroller is a high-performance embedded system widely used in electronic and automation applications. In this project, a 10 k Ω potentiometer is used as an analog input device, while three LEDs serve as output indicators. The potentiometer acts as a variable resistor that produces different voltage levels based on the position of its knob. These varying voltages are read by the STM32 through its Analog-to-Digital Converter (ADC), which converts the analog signal into a digital value that can be processed by the microcontroller.

Based on the ADC value obtained from the potentiometer, the STM32 controls the LEDs. Different voltage ranges correspond to different LED states. For example, when the potentiometer is at a low position, only the first LED may glow; at a medium position, the second LED may glow; and at a high position, the third LED may glow. This demonstrates the interfacing of analog sensors with digital outputs in embedded systems.

The LEDs are connected through 330 Ω resistors, which limit the current flowing through the LEDs and protect them from excessive current that could cause damage.

This project illustrates the concepts of analog signal acquisition, ADC operation, digital processing, GPIO control, and LED interfacing using the STM32 microcontroller. It provides a practical understanding of how microcontrollers interact with real-world analog inputs and generate corresponding outputs.

Interfaces Between Components

Sl. No.	Component / Module	Input	Output	Function
1	Potentiometer	Knob Position	Analog Voltage	Variable Voltage Source
2	STM32 ADC	Analog Voltage	Digital ADC Value	Analog-to-Digital Conversion
3	GPIO (Output)	ADC Digital Value	LED Drive Signal	Output Control
4	LED + Resistor	GPIO Drive Signal	Light Output	Visual Indicator

Simulink Model

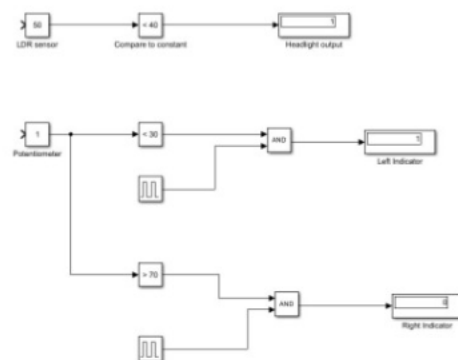


Figure 24: Simulink Model – Automatic Headlight and Indicator System

Breadboard Implementation

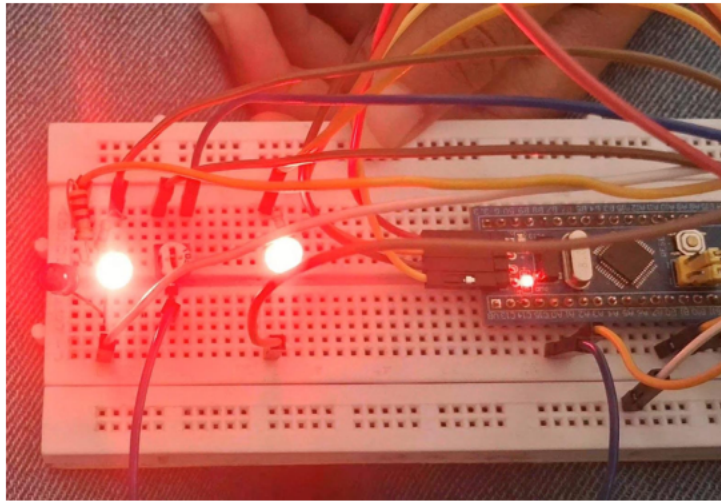


Figure 25: Breadboard Implementation – Automatic Headlight and Indicator System

Result and Conclusion

The STM32 microcontroller successfully read the analog voltage from the 10 k Ω potentiometer using its ADC module. Based on the potentiometer position, the ADC value changed continuously and the corresponding LEDs were controlled correctly. The three LEDs indicated different voltage ranges, demonstrating successful analog-to-digital conversion and GPIO output control. The 330 Ω resistors effectively limited the current through the LEDs, ensuring safe operation of the circuit. The system responded accurately and in real time to changes in the potentiometer position.

The project successfully demonstrated the interfacing of a 10 k Ω potentiometer and three LEDs with an STM32 microcontroller. The ADC module accurately converted the analog voltage from the potentiometer into digital values, and the LEDs responded according to the programmed thresholds. This experiment provided practical knowledge of ADC operation, GPIO programming, sensor interfacing, and output control in embedded systems. The results confirm that STM32 can efficiently process analog inputs and control digital outputs, making it suitable for various monitoring and automation applications.